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Clustering-augmented reversal strategy improves return performance: Evidence from Chinese stock market

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ABSTRACT

This paper proposes a novel clustering-augmented reversal strategy that employs the unsupervised K-means algorithm to group stocks before constructing contrarian portfolios. This strategy consistently outperforms the standard reversal approach in the Chinese stock market, generating significant monthly long-short returns. The performance gains are primarily driven by short positions, while long positions remain insignificant. Notably, technical characteristics prove more effective than fundamental characteristics in clustering stocks, largely due to the retail-dominated structure of the A-share market. Clustering proves crucial, contributing approximately 20 %–45 % of the total portfolio returns. Analysis of return quantile transition distributions further confirms that reversals are more pronounced among past winners than among losers. Finally, risk-adjusted regressions reveal that the long-short portfolios generate significantly positive alphas while exhibiting no significant loadings on risk factors, underscoring a risk-offsetting mechanism within clusters.

1. Introduction

The reversal in stock returns is a well-documented phenomenon in literature. Jegadeesh (1990) and Lehmann (1990) show that a contrarian strategy of buying loser stocks and selling winner stocks could yield economically significant returns. Previous literature has explained the source of reversals mainly in two ways. The first is the liquidity provision hypothesis, which posits that reversals represent compensation for liquidity providers bearing risk under adverse market conditions (Avramov et al., 2006; Campbell et al., 1993; Cheng et al., 2017; Hameed and Mian, 2015; Nagel, 2012). The second centers on behavioral factors, such as investor overreaction to firm-specific information, belief reversals, or investor heterogeneous beliefs (Cooper, 1999; Li et al., 2022; Subrahmanyam, 2005).

Although momentum and reversal have been observed in both the U.S. market and Chinese stock markets, they differ substantially in terms of profitability and duration. In the U.S. market, momentum strategies typically yield positive returns over short investment horizons (usually less than one year), while reversal strategies tend to be profitable over horizons longer than one year (Asness et al., 2013; Griffin et al., 2003; Jegadeesh and Titman, 1993; Levy, 1967). In contrast, Gang (2019) and Yu et al. (2019) show that in the Chinese market, momentum persists only over very short horizons (as brief as one week), whereas reversal strategies become profitable beyond that period. This discrepancy is primarily attributed to the higher proportion of individual investors in China compared to the U.S. (Gang, 2019; Yu et al., 2019). Aiming at this widely prevalent reversal phenomenon in the Chinese stock market, this paper

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proposes a clustering-augmented strategy to enhance returns.

Hameed and Mian (2015) argue that returns on firms within the same industry are highly contemporaneously correlated, as they are similarly exposed to common shocks. They suggest that if return reversals reflect deviations from fundamental values followed by subsequent corrections, then matching firms with similar fundamentals (or industries) should help better identify short-term return reversals. Consistent with this view, they classify stocks by industry and find that intra-industry reversals are more significant than the unconditional reversals, providing supportive evidence for the importance of clustering. Building on this insight, our paper proposes a clustering-augmented reversal strategy. Different from Hameed and Mian (2015)'s simple industry-based classification, we employ machine learning algorithms to cluster stocks based on their historical technical and fundamental characteristics. This approach allows us to form clusters of stocks with similar price dynamics and potential co-movements. Within each cluster, we identify the top 10 % winners and bottom 10 % losers during the formation period and construct long-short portfolios, which are then held for one month.

Liao et al. (2024) and Moosa and Li (2011) suggest that individual investors, the dominant investor group in the Chinese market, primarily rely on technical analysis while largely ignoring fundamental information. Motivated by this insight, we first cluster stocks using only technical characteristics over the past 3, 6, 9, and 12 months, and find that the long-short portfolios yield highly significant raw monthly returns ranging from 2.329 % to 2.515 % under the equal-weighted scheme, and from 1.838 % to 2.064 % under the value-weighted scheme. Notably, the majority of the long-short returns are driven by the short leg, while the long leg returns are small in magnitude and statistically insignificant. Next, we compare the portfolio performances when different sets of characteristics are used and find that technical characteristics appears to be more effective than fundamental ones in stock clustering, largely due to the retail-dominated structure of the A-share market.

We then assess the performance gains from clustering relative to the standard reversal strategy. We find that for any number of clusters greater than one, the long-short strategy consistently outperforms the standard reversal strategy (cluster = 1). For instance, under the 3-month clustering period and value-weighted scheme, the standard strategy yields a monthly return of 1.281 %, which, however, increases to 2.201 % when 13 clusters are used, implying that clustering contributes to approximately 42 % of the total return.

To further examine the presence of mean reversion within clusters, we analyze the probability distribution of return quantile transitions between the formation and holding periods. The results confirm that mean reversion occurs within clusters and is more pronounced among past winners than among losers. In addition, an industry-level performance analysis reveals that the long-short portfolios deliver significantly positive returns across most industries, validating the robustness of our strategy.

To assess whether the superior performance of our enhanced reversal strategy is driven by increased risk exposure or genuine alpha, we conduct a risk-adjusted return analysis using the Fama-French factor model. The results show that the factor loadings of the long-short portfolios on all three Fama-French factors are statistically insignificant, while the alphas remain significantly positive, ranging from 2.281 % to 2.505 % per month. These values are substantially higher than the 0.97 % reported by Hameed and Mian (2015) and 1.34 % reported by Da et al. (2014). The findings suggest that the long-short portfolios achieve positive abnormal returns without bearing increased exposure to systematic risk factors. Similar results are obtained using the Fama-French five-factor model.

Our contributions are reflected in the following aspects. First, we propose a novel clustering-augmented reversal strategy that significantly outperforms both the standard reversal strategy and the industry-classified reversal strategy, offering practical insights for investment applications. Second, by comparing monthly returns across different numbers of clusters, we demonstrate the importance of clustering, which contributes approximately 20 % to 45 % of the total returns. Third, we show that the long-short portfolios constructed under our strategy generate notable alphas while exhibiting no significant loadings on the Fama risk factors. This is because the long and short positions within each cluster share similar exposures to systematic risks, and the long-short structure effectively offsets these exposures.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the methodology and data. Section 4 presents the empirical results. Section 5 discusses the decomposition of portfolio returns, and Section 6 concludes.

2. Literature review

Jegadeesh (1990) finds that monthly stock returns exhibit significantly negative first-order serial correlation and shows that a reversal strategy of buying past losers and selling past winners could yield economically significant returns. Similarly, Lehmann (1990) reports that past-week winners (losers) tend to earn negative (positive) returns in the following week, and that the long-short profits persist after accounting for bid-ask spreads and plausible transactions costs. Moreover, the reversal phenomenon is not confined to equity markets. Prior research also documents significant price reversals in the bond market, showing that corporate bonds which underperformed in previous months are more likely to outperform those with higher past returns (Bali et al., 2021; H. Zhang and Wang, 2021).

The two most common explanations for the reversal phenomenon are the liquidity provision hypothesis and investor behavioral biases. The liquidity provision hypothesis posits that when stock prices decline, liquidity providers are willing to buy only in anticipation of subsequent price reversals and positive returns (Avramov et al., 2006; Campbell et al., 1993; Cheng et al., 2017; Hameed and Mian, 2015; Nagel, 2012). In this view, reversal represents compensation for liquidity providers bearing risk under adverse market conditions. Hameed et al. (2010) show that this compensation is particularly large during substantial market downturns, when the decreased collateral value of financial intermediaries and forced liquidations by asset holders make liquidity provision more difficult and costly. Furthermore, Avramov et al. (2006) find that the largest reversals and contrarian strategy profits are largely concentrated in small, high turnover, and illiquid stocks.

Another explanation attributes the reversal to investor behavior. For instance, Cooper (1999) suggests short-term return reversals are linked to investor overreaction to firm-specific information. Subrahmanyam (2005) argues that monthly reversals are largely driven by reversals in beliefs of financial market agents. Li et al. (2022) further show that heterogeneous investor beliefs can strengthen short-term reversal effects, with higher heterogeneity resulting in significantly lower returns for past winners and higher returns for past losers. However, Da et al. (2014) suggest that the two explanations are not necessarily mutually exclusive, whose research proves that both liquidity shocks and investor sentiment contribute to the short-term reversal, but in different sides. Specifically, they show that liquidity shocks mainly play a role in the long side because fire sales more likely demand liquidity, while investor sentiment works on the short side because short-sale constraints prevent the immediate elimination of overvaluation.

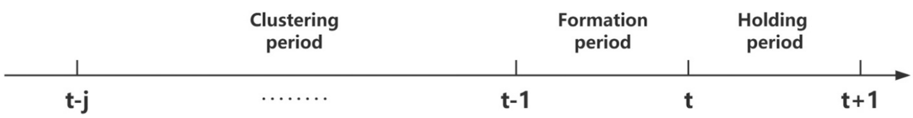
The reversal phenomenon in the Chinese stock market is more pronounced than in the U.S. market, largely due to its retail-dominated structure (Gang, 2019; Yu et al., 2019). Prior studies attribute stock reversals to retail investors' limited attention, overreaction, lottery-seeking preferences, and other irrational trading behaviors. Unlike institutional investors, who possess the resources to track a broad range of stocks, retail investors are constrained by limited attention and are more inclined to purchase attention-grabbing stocks (Barber and Odean, 2008). B. Zhang and Wang (2015) and Yung and Nafar (2017) use online search index to measure retail investor attention and find that increased attention generates positive price pressure, temporarily inflating prices above fundamental values before subsequent corrections, thereby inducing reversals. Welagedara et al. (2017) show that when negative news arrives, stocks with greater retail attention tend to overreact, exhibiting significantly stronger price reversals compared with those predominantly followed by institutional investors. George et al. (2023) further demonstrate that reversals intensify in February, as retail investors during the Chinese New Year period exhibit a heightened appetite for lottery-like stocks, often purchasing past losers in pursuit of high payoffs. Chen et al. (2025) also provide evidence that retail investors' preference for lottery-like payoffs amplifies their overreaction to news, resulting in larger short-term return reversals.

A large body of literature has also sought to enhance the performance of the standard reversal strategy. For instance, Zhu et al. (2021) show that price reference points matter, as reversal strategies based on a stock's distance from its 52-week high outperform those constructed using past returns. Hameed and Mian (2015) demonstrate that matching firms with similar fundamentals can more accurately capture short-term reversals. They achieve this by classifying stocks by industry and find that the magnitude of the intra-industry reversals exceeds the unconditional reversals, which highlights the significance of clustering.

3. Methodology and data

3.1. Methodology

In this section, we introduce our core methodology for capturing within-cluster reversals, which is structured around three main periods.



3.1.1. Clustering period ($t-j$ to $t-1$)

During the clustering period, we use an unsupervised machine learning algorithm to group stocks into clusters based on their technical and fundamental characteristics, in order that stocks exhibiting similar price trajectories and trading patterns would be classified into the same cluster. This approach enables each cluster to represent a group of stocks with similar historical behavior and underlying co-movements.

3.1.2. Formation period ($t-1$ to t)

Following stock clustering, we examine the performance of individual stocks within each cluster over the subsequent one-month formation period. Stocks are ranked within each cluster based on their returns during this period. The top 10 % of stocks with the highest returns are classified as winners, while the stocks in bottom 10 % are identified as losers. In this step, we aim to pick up those stocks who have deviated the most from their cluster average. We then construct contrarian portfolios by taking long positions in the losers and short positions in the winners within each cluster at time t.

3.1.3. Holding period (t to $t+1$)

After constructing the contrarian portfolio, we hold it for one month. Since each cluster consists of stocks with the most similar return patterns, we expect to see short-term mean reversion within clusters. Specifically, we posit that the past winner (loser) stocks are likely to experience price reversals and thus underperform (outperform) their cluster averages during the holding period, leading the contrarian portfolio to yield positive returns.

The entire process is implemented using a rolling-window approach, with clusters and portfolios updated monthly. By repeating this procedure, we aim to exploit intra-cluster reversals to enhance return performance.

3.2. K-means method

K-means, the clustering method employed in this study, is one of the most widely used unsupervised learning algorithms for partitioning a dataset into a predefined number of clusters, denoted by K. The objective of the K-means algorithm is to partition n observations into K clusters by minimizing the within-cluster sum of squared distances (WCSS). Each observation is assigned to the cluster with the nearest centroid, thereby maximizing intra-cluster similarity and inter-cluster dissimilarity.

Mathematically, the objective function is given by:

$$\min_{\{C_1,C_2,\ldots,C_K\}} \sum_{k=1}^{K} \sum_{x_i \in C_k} ||x_i - \mu_k||^2$$

where C_k denotes the set of observations assigned to cluster k, x_i is an individual data point, and $\mu_k = \frac{1}{|C_k|}\sum_{x_i \in C_k} x_i$ is the centroid of cluster k.

In the literature, K-means has been one of the most widely applied clustering algorithms for stock classification (Ansari, 2024; Das et al., 2017; Fang and Chiao, 2021; Wu et al., 2022). Wu et al. (2022) highlight that K-means is conceptually simple, easy to implement, computationally efficient, and yields robust clustering outcomes with strong interpretability. Similarly, Fang and Chiao (2021) emphasize that K-means demonstrates strong performance in data classification, facilitating more accurate stock prediction. These advantages motivate our choice of K-means as the baseline method.

Nonetheless, K-means is not without limitations: its results depend on the choice of initial values, which must be specified ex ante, and it is sensitive to outliers. Therefore, to address these drawbacks and ensure the robustness of our findings, we complement our analysis with alternative clustering techniques in the later section.

3.3. Data and variable

Our sample includes all A-share stocks listed on the Shanghai Stock Exchange (SSE) and the Shenzhen Stock Exchange (SZSE) from January 2010 to December 2024. The choice of January 2010 as the starting point is driven by the data requirements of the clustering methodology, which necessitates a sufficiently large cross-section of stocks to form robust clusters. Prior to 2010, the number of eligible stocks was relatively limited. From 2010 onward, however, the eligible stock count exceeded 1500, allowing for more reliable clustering and portfolio construction.

We use two categories of stock characteristics: technical characteristics and fundamental characteristics, both sourced from the CSMAR database. For technical features, we use returns and return volatility during the clustering period. For example, if the clustering window spans six months, then we calculate returns and volatility over the periods $[t-6,t]$, $[t-5,t]$, …, $[t-1,t]$, respectively. For fundamental features, we incorporate variables that capture key financial dimensions such as profitability, size, growth, and value. The specific indicators used are summarized in Table 1.

To ensure data quality and strategy robustness, we implement the following data filtering and processing criteria. First, we include only stocks that have been listed for more than one year at the time of portfolio formation, to ensure the availability of historical information. Second, in cases where a stock is delisted or suspended during the holding period, we use its last available valid price within the holding window to calculate returns. Third, if a cluster contains fewer than 20 constituent stocks, then the cluster is discarded to mitigate the risk of poor diversification and unstable return estimates due to small sample sizes.

4. Empirical results

4.1. Baseline results

In the baseline setting, we use only technical features during the clustering period to classify all stocks into 20 clusters.[1] Panel A of Table 2 reports the monthly raw returns of equal-weighted portfolios constructed based on this clustering approach. As shown, the long-short portfolios deliver significantly positive returns across all clustering window lengths, yielding average monthly raw returns of 2.424 %, 2.515 %, 2.470 %, and 2.329 % for 3-, 6-, 9-, and 12-month periods, respectively. All of these estimates are statistically significant at the 1 % level.

A closer inspection reveals that the majority of the long-short returns are driven by the short leg of the strategy. Specifically, the average monthly returns of the short portfolios are −1.998 %, −2.093 %, −2.195 %, and − 2.028 % for the 3-, 6-, 9-, and 12-month clustering periods, respectively, all significant at the 1 % level. In contrast, the long leg returns are small in magnitude and statistically insignificant across all clustering periods. This indicates that the profits are primarily attributable to those stocks that outperformed their cluster peers during the formation period but subsequently experienced price declines during the holding period.

Panel B of Table 2 presents the results for value-weighted portfolios. Similar to the equal-weighted results, the value-weighted long-short portfolios yield statistically significant returns of 1.948 %, 2.064 %, 2.047 %, and 1.838 % per month across the 3-, 6-, 9-, and 12-month clustering periods, respectively. Likewise, the return contribution predominantly comes from the short leg, while the long leg

[1] We will examine how monthly raw returns vary with the number of clusters in a later section.

Table 1
Summary of stock characteristics used.

Stock characteristics		Description
Technical characteristics	$Ret_{t-i,t}$	Return from month $t-i$ to t
	$Std_{t-i,t}$	Standard deviation of daily returns from month $t-i$ to t
	MktValue	Market capitalization of firms
	NetProfit	Net profit in the most recent fiscal quarter
	ProfitMargin	Net profit divided by total revenue
	ROE	Net income divided by shareholders' equity
	ROA	Net income divided by total assets
Fundamental characteristics	*EPS*	Earnings per share in the most recent fiscal quarter
	AssetGrowth	Year-over-year percentage change in total assets
	ProfitGrowth	Year-over-year percentage change in net profit
	RevenueGrowth	Year-over-year percentage change in total revenue
	PE	Market price divided by earnings per share
	BM	Book value divided by market value

Notes: This table summarizes the stock-level characteristics used as input variables in the clustering process.

Table 2
Monthly raw returns of portfolios under different clustering periods.

Panel A: Monthly raw returns of equal-weighted portfolios with 20 clusters				
	Clustering Period			
	3 months	6 months	9 months	12 months
Long Return	0.426	0.421	0.275	0.301
	(0.66)	(0.65)	(0.43)	(0.45)
Short Return	−1.998***	−2.093***	−2.195***	−2.028***
	(−3.28)	(−3.47)	(−3.56)	(−3.25)
Long-Short Return	2.424***	2.515***	2.470***	2.329***
	(6.92)	(6.86)	(7.15)	(6.36)
Panel B: Monthly raw returns of value-weighted portfolios with 20 clusters				
	Clustering Period			
	3 months	6 months	9 months	12 months
Long Return	0.153	0.241	0.081	0.055
	(0.25)	(0.39)	(0.13)	(0.09)
Short Return	−1.795***	−1.823***	−1.966***	−1.783***
	(−3.07)	(−3.20)	(−3.35)	(−3.01)
Long-Short Return	1.948***	2.064***	2.047***	1.838***
	(4.92)	(5.19)	(5.20)	(4.44)

Notes: This table reports the monthly raw returns (in percentages) of long, short, and long-short portfolios. Panel A presents the results for equal-weighted portfolios, and Panel B reports the results for value-weighted portfolios. The long portfolio consists of the loser stocks (bottom 10 %) within each cluster during the formation period, while the short portfolio consists of the winner stocks (top 10 %). The long-short portfolio represents a zero-cost strategy that longs losers and shorts winners within each cluster. The reported t-statistics are in parentheses. *, **, and *** indicate statistical significance at the 10 %, 5 %, and 1 % levels, respectively.

remains insignificant.

Notably, for any given clustering period, the long-short returns under equal weighting are consistently larger than those under value weighting. This pattern suggests that the return contributors are mainly small-cap stocks.

In addition, we also consider portfolio returns with 10, 15, 25, and 30 clusters, and the corresponding results are reported in Appendix A. The results indicate that varying the number of clusters does not alter the key findings: the long-short portfolio returns remain highly significant, and the structural characteristics are consistent with those obtained under the 20-cluster specification. This evidence confirms the robustness of our strategy.

We report in detail the standard deviations of the portfolios' monthly raw returns, discuss the strategy's risk profile, and display the cumulative returns of the long-short portfolios in Appendix B for readers' reference.

4.2. The comparison of performance across different stock characteristics

The effectiveness of fundamental-based investment in China has long been a subject of debate. Institutional investors are widely recognized as sophisticated market participants, capable of perceiving relevant fundamentals, analyzing firm-related information, and enhancing market efficiency (Akbas et al., 2015; Y. Chen et al., 2019; Kokkonen and Suominen, 2015; Paiardini, 2015). In contrast, retail investors are generally less informed and weaker in analyzing the information (Liao et al., 2024), display greater dispersion in beliefs (Bian et al., 2023), tend to follow market trends rather than fundamentals (Barber and Odean, 2008).

Given that the Chinese stock market is dominated by retail investors, it has frequently been criticized as speculative in nature and as lacking a strong linkage between equity valuation and firm quality (Fong and Toh, 2014; Gu et al., 2018; Ng and Wu, 2007). Moosa and Li (2011) provides mixed results regarding the effectiveness of fundamental trading but find strong support for technical trading in China. They explain this phenomenon from several perspectives. First, the government plays an important role in the Chinese market, holding substantial stakes in many listed firms. Since it often uses the market to achieve policy objectives such as credibility and stability, stock prices may diverge from economic fundamentals. Second, the market is dominated by retail investors who lack professional expertise and behave as “noise traders”, further widening the gap between prices and intrinsic values. Third, weak protection for small investors’ interest fosters short-term speculation rather than long-term fundamental investment. Against this backdrop, this section aims to compare strategy performances when different sets of characteristics are used to cluster stocks.

Table 3 and Table 4 present the monthly long-short returns of equal-weighted and value-weighted portfolios, respectively, constructed on different sets of characteristics. Panel A of both tables compares the performance of portfolios clustered by technical characteristics, fundamental characteristics, and their combination. The results indicate that technical information consistently delivers higher portfolio returns than fundamentals. Interestingly, combining technical and fundamental characteristics does not enhance performance relative to using technical information alone. This finding suggests that adding more information does not necessarily improve performance, as redundant or noisy signals may dilute effective trading signals.

Panel B of the two tables further investigates whether different subsets of fundamentals, including size, value, profitability, and growth, exhibit heterogeneous effects on performance. The results reveal clear heterogeneity across different dimensions of fundamentals. For equal-weighted portfolios, value and growth characteristics appear to generate stronger returns than size or profitability, whereas in value-weighted portfolios, the profitability and value dimensions play a more prominent role.

Overall, the evidence suggests that technical characteristics appears to be more effective than fundamental ones in the Chinese A-share market. The combination of technical and fundamental signals does not necessarily improve performance, as more information may increase noise rather than enhance predictability. In addition, the heterogeneity analysis underscores that different dimensions of firm fundamentals contribute unevenly, and their relative effectiveness depends on portfolio construction methods.

4.3. Performance improvements attributable to clustering

In this section, we compare our clustering-augmented reversal strategy with the standard reversal strategy to evaluate the performance gains from incorporating the clustering step. The standard reversal strategy ranks stocks solely by past returns and constructs a long-short portfolio by buying past losers and selling past winners. In contrast, our approach introduces a clustering layer that groups similar stocks before applying the contrarian logic within each cluster.

Fig. 1 illustrates how the monthly returns of the long-short portfolio vary with the number of clusters under both equal-weighted and value-weighted schemes across different clustering horizons (3, 6, 9, and 12 months). The red points represent the portfolio where the number of clusters equals one, corresponding to the standard reversal strategy without prior segmentation. The green points mark the highest return achieved along each curve, representing the optimal performance of the clustering-augmented strategy.

Fig. 1 clearly demonstrates that introducing the clustering step significantly enhances portfolio performance. For instance, under the 3-month clustering horizon (top-left panel), the equal-weighted strategy yields a monthly return of 2.107 % under the standard approach (cluster = 1), which, however, increases to 2.634 % when 13 clusters are used, indicating that the clustering step contributes to around 20 % of the total returns. The value-weighted counterpart improves even more dramatically, from 1.281 % to 2.201 %, reflecting a return contribution ratio of about 42 %. Similar performance gains are observed across other clustering horizons and weighting schemes.

Notably, for any number of clusters greater than one, the long-short strategy consistently outperforms the standard reversal benchmark (cluster = 1). This suggests that the benefits of clustering are not confined to specific cluster counts, underscoring the robustness of our approach. By grouping stocks with similar return characteristics prior to portfolio construction, the strategy is better able to exploit relative mispricing and local mean reversion, thus generating higher and more stable returns.

4.4. Evidence of within-cluster mean reversion

We posit that the profitability of our clustering-augmented reversal strategy is primarily driven by the mean-reversion behavior of stocks within each cluster. Specifically, we expect that stocks which deviate significantly from their intra-cluster average return during the formation period will tend to revert toward the cluster mean in the holding period. In other words, we expect formation-period winners to underperform and losers to outperform in the holding period as they revert toward the intra-cluster mean.

To empirically examine the presence of mean reversion within clusters, we analyze the probability distribution of return quantile transitions between the formation and holding periods. Figs. 2–5 present the transition matrices for stocks with clustering periods of 3, 6, 9, and 12 months, respectively. In each matrix, stocks are sorted into deciles based on their returns in the formation period (rows) and then re-ranked into deciles based on their returns in the holding period (columns). Each cell represents the conditional probability that a stock transitions from a given formation-period decile to a given holding-period decile.[2]

A consistent pattern emerges across all four figures: stocks in the top decile (0–10 %) during the formation period exhibit a

[2] By definition, each row sums to one.

Table 3
Monthly long-short returns of equal-weighted portfolio with different characteristics used to cluster stocks.

Panel A: Monthly long-short returns of equal-weighted portfolio using technical and fundamental characteristics				
	Clustering Period			
	3 months	6 months	9 months	12 months
Technical	2.424***	2.515***	2.470***	2.329***
	(6.92)	(6.86)	(7.15)	(6.36)
Fundamentals	1.991***	1.957***	1.991***	1.892***
	(5.52)	(5.35)	(5.37)	(5.07)
Technical & Fundamentals	2.075***	2.078***	2.063***	2.145***
	(5.99)	(5.67)	(5.86)	(5.83)
Panel B: Monthly long-short returns of equal-weighted portfolio using different subsets of fundamentals				
	Clustering Period			
	3 months	6 months	9 months	12 months
Size	1.467***	1.421***	1.456***	1.310***
	(3.90)	(3.73)	(3.79)	(3.44)
Value	2.018***	1.987***	2.043***	1.974***
	(6.33)	(6.15)	(6.26)	(6.02)
Profitability	1.819***	1.807***	1.855***	1.754***
	(4.83)	(4.72)	(4.78)	(4.51)
Growth	2.043***	2.028***	2.086***	2.063***
	(5.71)	(5.60)	(5.70)	(5.58)

Notes: This table reports the monthly long-short returns (in percentages) of equal-weighted portfolios constructed on different sets of characteristics. Panel A presents results for portfolios clustered using technical characteristics, fundamental characteristics, and their combination. Panel B shows results for portfolios clustered using different subsets of fundamentals, including size, value, profitability, and growth. The reported t-statistics are in parentheses. *, **, and *** indicate statistical significance at the 10 %, 5 %, and 1 % levels, respectively.

Table 4
Monthly long-short returns of value-weighted portfolio with different characteristics used to cluster stocks.

Panel A: Monthly long-short returns of value-weighted portfolio using technical and fundamental characteristics				
	Clustering Period			
	3 months	6 months	9 months	12 months
Technical	1.948***	2.064***	2.047***	1.838***
	(4.92)	(5.19)	(5.20)	(4.44)
Fundamentals	1.754***	1.738***	1.753***	1.617***
	(4.17)	(4.09)	(4.07)	(3.75)
Technical & Fundamentals	1.779***	1.787***	1.689***	1.804***
	(4.67)	(4.36)	(4.19)	(4.53)
Panel B: Monthly long-short returns of value-weighted portfolio using different subsets of fundamentals				
	Clustering Period			
	3 months	6 months	9 months	12 months
Size	1.356***	1.325***	1.366***	1.242***
	(3.50)	(3.36)	(3.43)	(3.12)
Value	1.424***	1.399***	1.437***	1.332***
	(3.92)	(3.80)	(3.85)	(3.56)
Profitability	1.457***	1.459***	1.500***	1.363***
	(3.25)	(3.20)	(3.24)	(2.96)
Growth	1.373***	1.391***	1.429***	1.386***
	(3.18)	(3.19)	(3.25)	(3.12)

Notes: This table reports the monthly long-short returns (in percentages) of value-weighted portfolios constructed on different sets of characteristics. Panel A presents results for portfolios clustered using technical characteristics, fundamental characteristics, and their combination. Panel B shows results for portfolios clustered using different subsets of fundamentals, including size, value, profitability, and growth. The reported t-statistics are in parentheses. *, **, and *** indicate statistical significance at the 10 %, 5 %, and 1 % levels, respectively.

significantly elevated probability of falling into the bottom decile (90–100 %) during the holding period. Specifically, this reversal probability is 24.11 %, 24.20 %, 24.24 %, and 24.26 % for the 3-, 6-, 9-, and 12-month clustering periods, respectively. These values are markedly higher than the unconditional probability of 10 % and provide compelling evidence of short-term reversal behavior among past outperformers.

In contrast, the behavior of bottom-decile stocks during the formation period is less clear-cut. The last row of each matrix, which corresponds to the worst-performing 10 % stocks during the formation period, shows a relatively uniform distribution across all holding-period deciles, implying that these losers do not exhibit significant reversal. This asymmetry indicates that while formation-

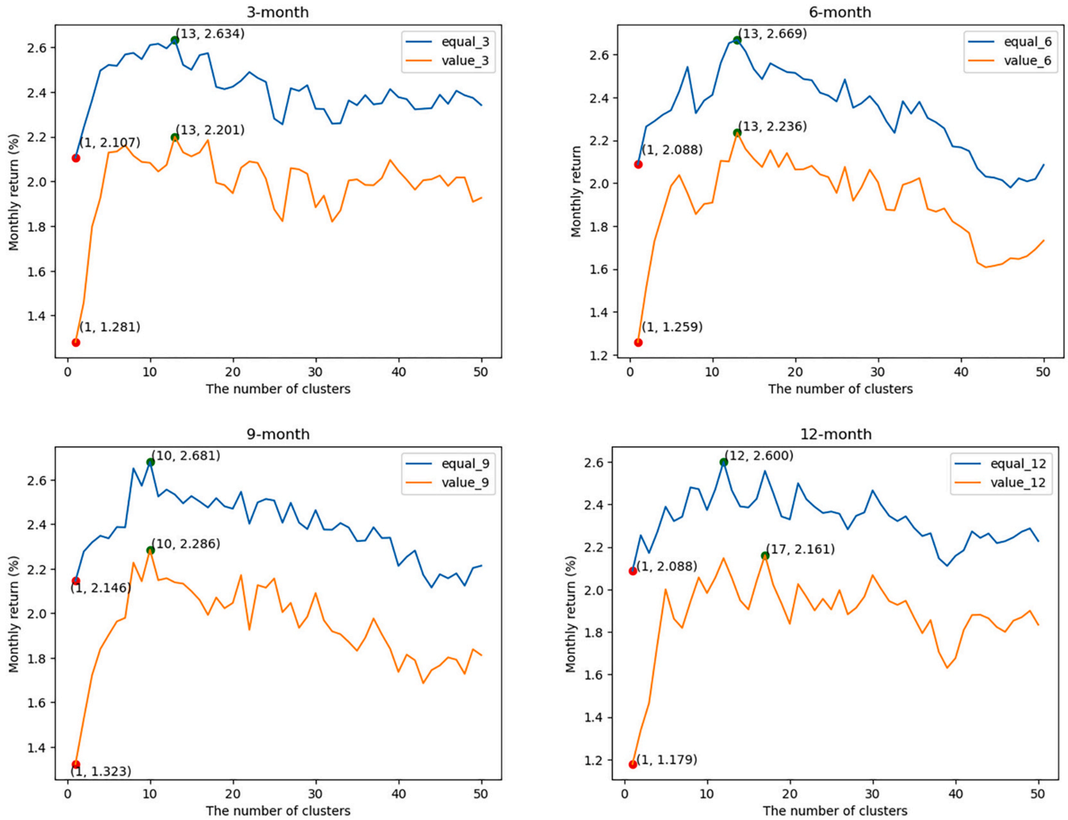


Fig. 1. Monthly returns across different numbers of clusters.
Notes: This figure presents the relationship between the number of clusters and the monthly returns of long-short portfolios under different clustering periods (3, 6, 9, and 12 months). Each panel compares the results of equal-weighted and value-weighted schemes. The x-axis represents the number of clusters used in the K-means algorithm, while the y-axis indicates the monthly returns of the corresponding long-short portfolio. The red dots mark the returns when no clustering is applied (i.e., the number of clusters equals one), corresponding to the standard reversal strategy. The green dots indicate the highest return observed along each curve, which reflects the optimal number of clusters for maximizing strategy performance.

period winners tend to mean-revert downward, formation-period losers do not significantly rebound.

Taken together, these findings provide micro-level support for our earlier results showing that the profits of the long-short strategy are primarily driven by the short leg. The transition matrices offer a direct confirmation of the within-cluster mean-reversion, while also highlighting the asymmetric nature of the reversal effect: mean reversion appears to be more pronounced among past winners than among losers.

4.5. Industry-level performance

To further investigate the robustness and heterogeneity of the enhanced reversal strategy, we conduct an industry-level performance analysis. We adopt the 2021 Shenwan Industry Classification Standard, which categorizes listed firms into 31 first-level industry sectors. The number of constituent stocks in each industry is reported in Table C1 in Appendix C. Since our strategy requires clustering stocks within each industry, a sufficiently large sample is necessary to ensure reliable grouping. Therefore, we restrict the analysis to sectors with more than 80 constituent stocks, resulting in a sample of 23 industries.[3]

Table C2 in Appendix C presents the monthly raw returns of the long, short, and long-short portfolios for these 23 industries under both equal-weighted and value-weighted schemes. The results show that the long-short portfolios deliver significantly positive returns across most industries, confirming that the reversal effect is pervasive across industries. Consistent with earlier findings, the return contribution is primarily driven by the short leg: in nearly all cases, the short portfolios yield statistically significant negative returns, whereas the long portfolios are generally insignificant.

[3] Given the limited number of stocks within each industry, we set the number of clusters to three in this industry-level analysis.

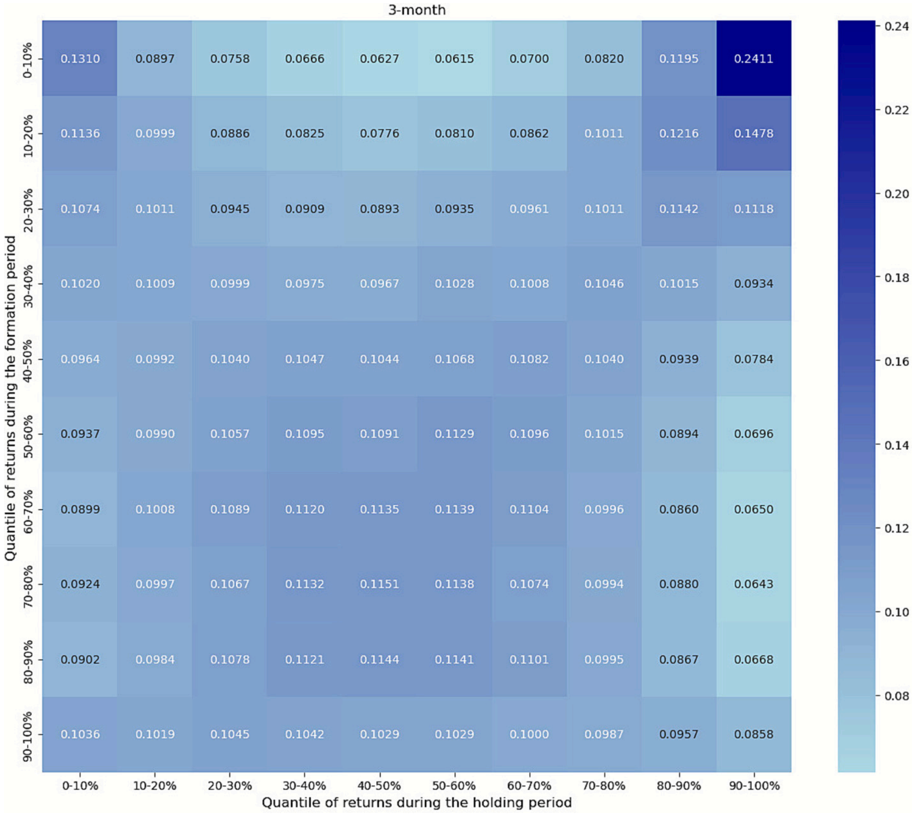


Fig. 2. Probability distribution of return quantile transitions with 3-month clustering period.
Notes: This heatmap illustrates the transition probabilities of stocks between return deciles when the clustering period is set to 3 months.

4.6. *Robustness checks*[4]

4.6.1. *Bid–ask effects*

Conrad et al. (1997), Jegadeesh and Titman (1995), and Kaul and Nimalendran (1990) document that a substantial portion of short-term reversal profits arises from negative serial covariance induced by bid-ask errors in transaction prices, and show that once the bid-ask bounce is accounted for, most profits are eliminated. Therefore, to examine whether our strategy remains valid after controlling for bid-ask effects, we recompute the monthly raw returns after skipping 1 day between the formation period and the holding period, following the approach of Hameed and Mian (2015).

Table E1 in Appendix E reports the results after excluding the first trading day of the holding period. The long-short monthly returns remain economically significant across different clustering horizons and weighting schemes, indicating that our strategy profits reflect more than the bid-ask bounce.

4.6.2. *Performance across different sizes*

Prior studies document that the strongest price reversals are concentrated in small-cap and illiquid stocks (Avramov et al., 2006; Hameed and Mian, 2015). To examine whether our clustering-augmented reversal strategy performs consistently across different firm sizes, we partition the full sample into two groups based on market capitalization. Specifically, at the beginning of each formation period, stocks are sorted by their most recent market value, with the top 50 % classified as the large-size group and the bottom 50 % as the small-size group.

Table E2 in Appendix E reports the monthly raw returns of long-short portfolios formed within each size group.[5] The results indicate that reversal profits are substantially stronger among small-cap stocks. Specifically, the magnitude of monthly returns in the small-size group is consistently more than twice that of the large-size group across all clustering horizons. Nevertheless, the large-size group still delivers statistically significant reversal profits, indicating that the strategy remains effective even among larger and more liquid stocks.

[4] We present all robustness check results in Appendix E.
[5] For brevity, only the long-short portfolio returns are reported here. Additional results are available from the authors upon request.

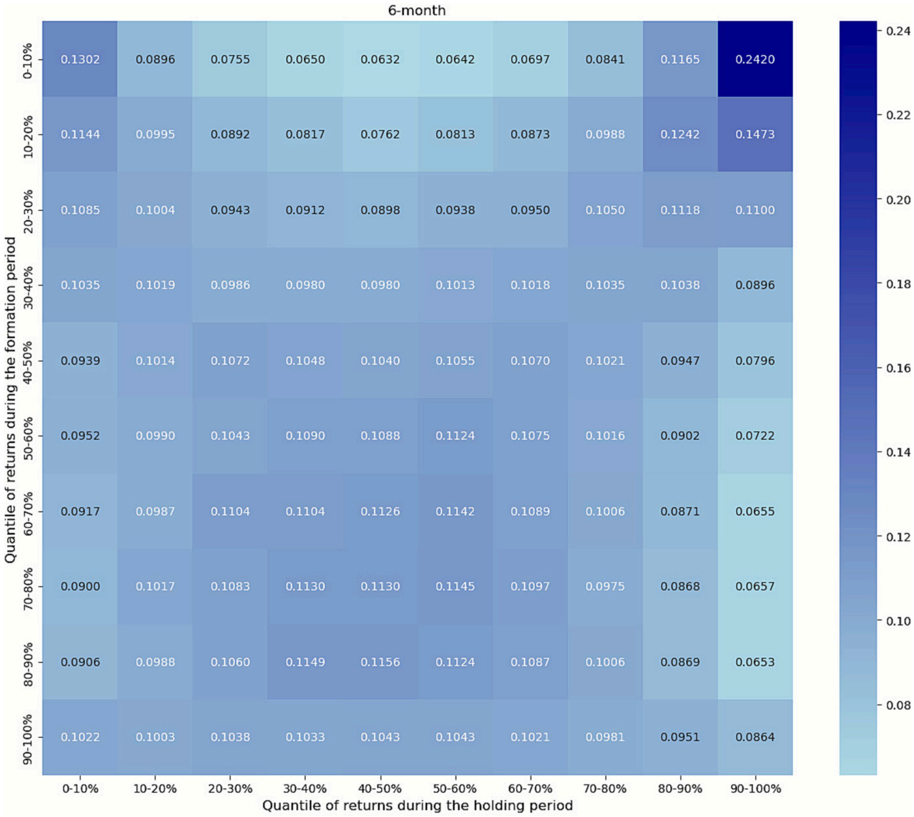


Fig. 3. Probability distribution of return quantile transitions with 6-month clustering period.
Notes: This heatmap illustrates the transition probabilities of stocks between return deciles when the clustering period is set to 6 months.

4.6.3. Portfolio returns under different clustering techniques

In the baseline analysis, we employ the K-Means algorithm to cluster stocks into groups. Prior studies have also adopted alternative clustering approaches in stock market applications, such as K-Medoids (Bayaraa et al., 2019; Gubu et al., 2024) and Hierarchical Clustering (Lestari et al., 2025; Teh and Cheong, 2015). Compared with K-Means, K-Medoids selects actual observations as cluster centers (medoids) rather than arithmetic means, making it more robust to outliers and noise, while Hierarchical Clustering constructs a nested hierarchy of clusters without requiring the number of clusters to be specified ex ante, thereby offering greater flexibility in uncovering data structure. We provide a detailed description of these two methods in Appendix D for readers interested in the methodological foundations.

In this section, we re-implement the strategy using K-Medoids and Hierarchical Clustering for the portfolio construction. The results, reported in Table E3 and Table E4, demonstrate that the long-short portfolio returns remain highly significant at the 1 % level under both alternative clustering methods, thereby confirming the robustness of our strategy to the choice of clustering algorithm.

4.6.4. Considering transaction costs

In practice, transaction costs inevitably erode a portion of trading profits. To evaluate the practical applicability of our strategy, this section analyzes portfolio performance net of transaction costs.

Total transaction costs consist of both explicit fees and implicit liquidity costs. Explicit costs primarily include commission and stamp tax. In the Chinese stock market, commission fees for institutional investors were around 0.05 % in 2012 but have since declined substantially. In recent years, commissions have generally ranged from 0.02 % to 0.03 % for retail investors and are even lower for institutional investors (Leippold et al., 2022). The stamp tax has been set at 0.1 % since 2008 and is levied unilaterally on sellers. Implicit liquidity costs arise mainly from bid-ask spreads and price slippage. According to Leippold et al. (2022), liquidity costs are typically around 0.1 % in the Chinese stock market.

Based on the above information, the total transaction cost is estimated at approximately 0.12 % to 0.15 % for buyers and 0.22 % to 0.25 % for sellers. This implies an average transaction cost of around 0.2 %, which Leippold et al. (2022) also identify as a realistic assumption. To further ensure robustness, we adopt a conservative approach by considering transaction costs of 0.2 %, 0.3 %, and 0.4 %, respectively.

Table E5 in Appendix E reports the monthly long-short portfolio returns after accounting for different levels of transaction costs. The results show a clear pattern in which returns gradually decline as transaction costs increase, revealing the erosive effect of trading

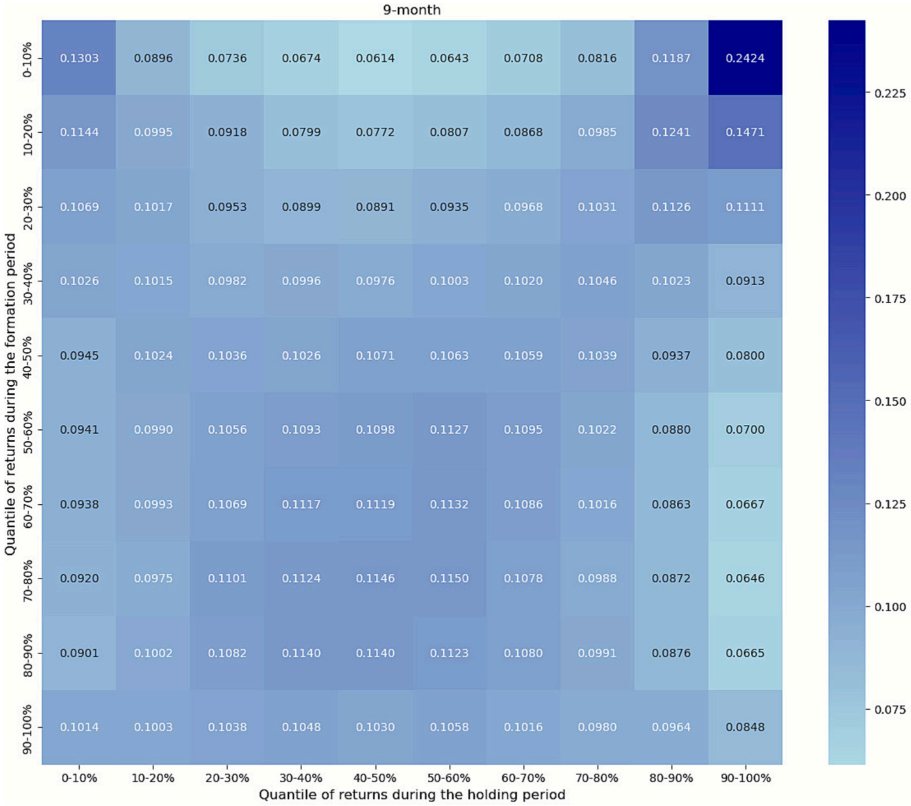


Fig. 4. Probability distribution of return quantile transitions with 9-month clustering period.
Notes: This heatmap illustrates the transition probabilities of stocks between return deciles when the clustering period is set to 9 months.

frictions on strategy returns. Nevertheless, even at the relatively high transaction cost of 0.4 %, the strategy continues to generate statistically significant net returns across all clustering periods and schemes. These findings provide evidence that our strategy's profitability remains robust in the presence of transaction costs.

4.6.5. Considering short-sale restrictions

The core profitability of our strategy lies in constructing long-short portfolios within each cluster. However, in the Chinese A-share market, short-sale is subject to restrictions, as not all stocks are eligible for shorting; only those included in the list of securities permitted for margin trading can be sold short. To incorporate this practical constraint, we refine our portfolio construction by restricting the short leg to eligible securities only. Moreover, to mitigate the risk of excessive concentration, we exclude clusters in which the number of stocks eligible for short selling falls below three.

Table E6 in Appendix E presents the monthly long-short portfolio returns after accounting for short-sale restrictions. Panel A reports the results when only short-sale restrictions are considered, while Panel B accounts for both short-sale restrictions and transaction costs. As shown in Panel A, portfolio returns decline relative to the baseline results once short-sale constraints are considered, suggesting that some stocks in the A-share market are indeed overpriced, but their subsequent correction cannot be exploited due to the inability to short sell. Nevertheless, the long-short portfolios still yield statistically significant returns, indicating that the strategy remains effective even under such restrictions. Panel B, which reflects a more realistic trading environment by considering both transaction costs and short-sale restrictions, shows that all portfolio returns still remain significant. This evidence highlights the robustness and practical applicability of our strategy in real investment settings.

5. Factor-based decomposition of portfolio returns

To evaluate whether the superior performance of our enhanced reversal strategy is driven by increased risk exposure or genuine alpha, we conduct a risk-adjusted return analysis based on the Fama-French factor model. The factor data for the Chinese stock market are sourced from the CSMAR database.

5.1. Decomposition based on Fama-French three factor model

Table 5 reports the regression results of the equal-weighted portfolios on the Fama-French three factors. The long portfolios exhibit

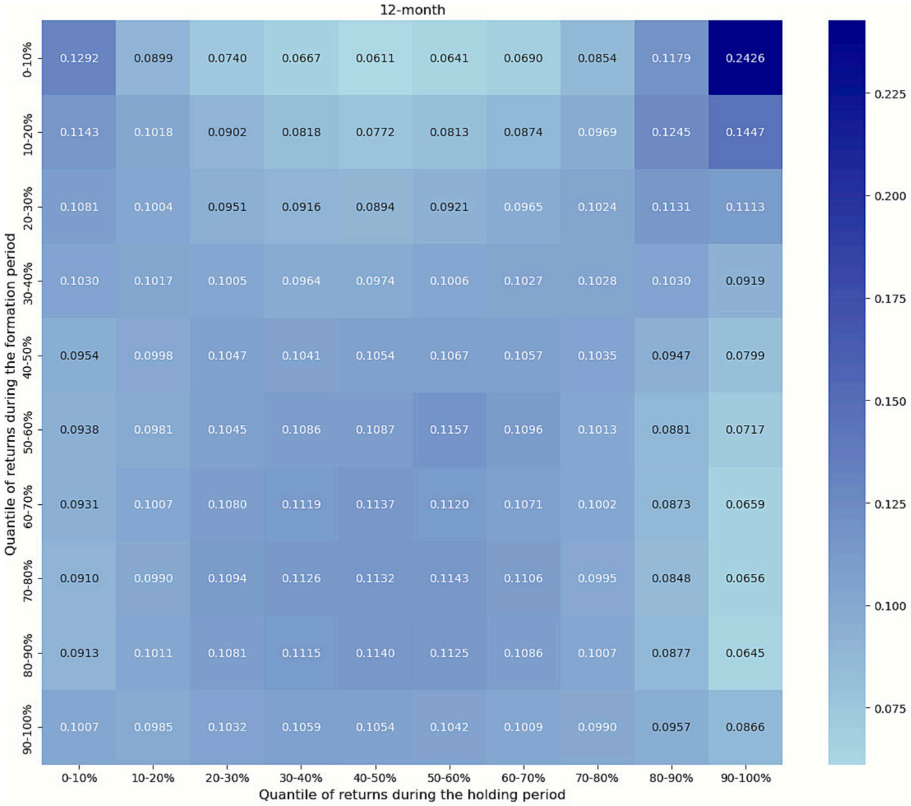


Fig. 5. Probability distribution of return quantile transitions with 12-month clustering period.
Notes: This heatmap illustrates the transition probabilities of stocks between return deciles when the clustering period is set to 12 months.

significant loadings on all three factors, with positive exposures to the market and size factors and negative exposures to the value factor. However, their alphas are statistically insignificant across all clustering horizons, suggesting that the observed returns of the long leg can be largely explained by risk factor exposures. The short portfolios exhibit a broadly similar pattern of factor loadings to the long portfolios, with positive exposures to the market and size factors and negative exposures to the value factor. Crucially, their alphas are significantly negative across all clustering periods, indicating that the underperformance of the short leg cannot be fully attributed to common risk factors but instead reflects a genuine reversal effect.

Interestingly, for the long-short portfolios, the factor loadings on the three Fama-French factors are almost all statistically insignificant, while the alphas remain significantly positive across all clustering horizons, ranging from 2.281 % to 2.505 % per month. This indicates that the long-short portfolios generate positive abnormal returns without bearing significant exposure to systematic risk factors. Moreover, relative to the risk-adjusted monthly return of 0.97 % reported by Hameed and Mian (2015) and 1.34 % reported by Da et al. (2014), our strategy delivers substantially higher returns.

The absence of factor exposures in the long-short portfolios underscores a risk-offsetting mechanism within clusters. Since the long and short portfolios exhibit similar exposures to the common risk factors, combining them in a long-short structure effectively cancels out those risks. These results underscore the strength of our cluster-based enhancement. By grouping stocks with similar return characteristics, the strategy naturally forms clusters in which stocks load similarly on risk factors. Consequently, constructing long-short positions within clusters allows us to exploit temporary mispricing while simultaneously hedging out common factor risks.

Table 6 reports the regression results for the value-weighted portfolios. The findings are broadly consistent with those of the equal-weighted case, further reinforcing the robustness of our strategy.

5.2. Decomposition based on Fama-French five-factor model

To further investigate whether our strategy's profitability is attributable to exposure to additional systematic risk factors, we extend the risk-adjusted return analysis to the Fama-French five-factor model. The regression results for the equal-weighted and value-weighted portfolios are reported in Tables F1 and F2 in Appendix F, respectively.

The findings closely resemble those from the earlier three-factor regressions. Both the long and short portfolios load positively on the market and size factors and negatively on the value factor, while their exposures to the profitability and investment factors are statistically insignificant. For the long-short portfolios, we find no significant loadings on any of the five factors, whereas the alphas remain positive and statistically significant across all clustering horizons.

Table 5
Risk-adjusted returns of equal-weighted portfolios based on the Fama-French three-factor model.

		Clustering Period			
		3 months	6 months	9 months	12 months
Long	α(%)	−0.276	−0.286	−0.316	−0.367
		(−0.97)	(−0.97)	(−1.11)	(−1.27)
	Mkt	0.883***	0.871***	0.860***	0.859***
		(17.98)	(16.81)	(17.09)	(16.97)
	SMB	0.665***	0.652***	0.638***	0.685***
		(9.11)	(8.63)	(8.76)	(9.38)
	HML	−0.363***	−0.399***	−0.410***	−0.387***
		(−3.47)	(−3.69)	(−3.91)	(−3.66)
Short	α(%)	−2.657***	−2.792***	−2.801***	−2.648***
		(−8.39)	(−8.61)	(−8.69)	(−8.01)
	Mkt	0.850***	0.846***	0.839***	0.838***
		(15.48)	(14.85)	(14.75)	(14.44)
	SMB	0.592***	0.608***	0.618***	0.577***
		(7.25)	(7.33)	(7.51)	(6.90)
	HML	−0.227*	−0.159	−0.220*	−0.255**
		(−1.94)	(−1.34)	(−1.86)	(−2.10)
Long-Short	α(%)	2.381***	2.505***	2.485***	2.281***
		(6.73)	(6.82)	(7.10)	(6.19)
	Mkt	0.032	0.026	0.022	0.021
		(0.53)	(0.40)	(0.35)	(0.33)
	SMB	0.073	0.043	0.020	0.107
		(0.80)	(0.46)	(0.22)	(1.15)
	HML	−0.136	−0.240*	−0.190	−0.132
		(−1.04)	(−1.78)	(−1.47)	(−0.98)

Notes: This table presents the regression results of monthly returns of equal-weighted portfolio on the Fama-French three factors: market (*Mkt*), size (*SMB*), and value (*HML*). The regressions are performed separately for the long, short, and long-short portfolios across different clustering periods (3, 6, 9, and 12 months). α is expressed in percentage terms. Figures in parentheses are t-statistics. *, **, and *** denote statistical significance at the 10 %, 5 %, and 1 % levels, respectively.

Table 6
Risk-adjusted returns of value-weighted portfolios based on the Fama-French three-factor model.

		Clustering Period			
		3 months	6 months	9 months	12 months
Long	α(%)	−0.381	−0.314	−0.344	−0.453
		(−1.28)	(−1.05)	(−1.16)	(−1.48)
	Mkt	0.912***	0.918***	0.911***	0.896***
		(17.71)	(17.43)	(17.35)	(16.67)
	SMB	0.390***	0.387***	0.368***	0.420***
		(5.10)	(5.04)	(4.85)	(5.43)
	HML	−0.518***	−0.523***	−0.561***	−0.518***
		(−4.73)	(−4.76)	(−5.12)	(−4.62)
Short	α(%)	−2.269***	−2.347***	−2.379***	−2.241***
		(−6.70)	(−7.18)	(−6.94)	(−6.56)
	Mkt	0.907***	0.888***	0.889***	0.882***
		(15.44)	(15.46)	(14.71)	(14.70)
	SMB	0.260***	0.303***	0.283***	0.285***
		(2.98)	(3.62)	(3.23)	(3.30)
	HML	−0.311**	−0.249**	−0.310**	−0.322**
		(−2.49)	(−2.08)	(−2.46)	(−2.57)
Long-Short	α(%)	1.888***	2.033***	2.035***	1.788***
		(4.80)	(5.16)	(5.18)	(4.33)
	Mkt	0.005	0.030	0.022	0.014
		(0.07)	(0.43)	(0.32)	(0.19)
	SMB	0.130	0.084	0.085	0.135
		(1.28)	(0.83)	(0.85)	(1.29)
	HML	−0.207	−0.274*	−0.251*	−0.196
		(−1.42)	(−1.90)	(−1.73)	(−1.30)

Notes: This table presents the regression results of monthly returns of value-weighted portfolio on the Fama-French three factors: market (*Mkt*), size (*SMB*), and value (*HML*). α is expressed in percentage terms. Figures in parentheses are t-statistics. *, **, and *** denote statistical significance at the 10 %, 5 %, and 1 % levels, respectively.

Taken together, these results confirm that the long-short portfolio returns are not compensation for bearing systematic risks captured by either the three- or five-factor models. Instead, they reflect idiosyncratic mispricing within clusters, which our method efficiently identifies and exploits.

6. Conclusion

This paper proposes a clustering-augmented reversal strategy which significantly improves return performance relative to the standard reversal strategy. Building on prior evidence that matching firms with similar characteristics can help better identify short-term return reversals, we apply the unsupervised K-means algorithm to cluster stocks based on historical stock-level features before constructing contrarian portfolios. By exploiting within-cluster reversals, the strategy achieves superior returns, yielding monthly returns of up to 2.515 % (equal-weighted) and 2.064 % (value-weighted). These gains are primarily driven by the short positions, indicating that overpriced stocks tend to revert more strongly. Interestingly, technical characteristics prove more effective than fundamental ones in clustering stocks, largely due to the retail-dominated structure of the Chinese stock market.

Further analyses demonstrate that the clustering-augmented strategy consistently outperforms the standard reversal strategy that does not segment stocks, with clustering contributing approximately 20 %–45 % of the total portfolio returns. Transition probability analysis offers a direct confirmation of the presence of within-cluster mean-reversion and also reveals that reversals are more pronounced among past winners than among losers.

Importantly, the enhanced returns cannot be explained by increased exposure to systematic risks. Risk-adjusted regressions using the Fama-French three- and five-factor models show that the long-short portfolios generate significantly positive alphas without significant loadings on risk factors. This suggests that the excess returns of our strategy are not compensation for risk but represent genuine alphas.

Overall, our findings provide robust evidence that clustering enhances reversal strategies by more effectively capturing short-term mispricing. The approach offers practical implications for asset managers seeking to improve contrarian trading performance.

CRediT authorship contribution statement

Weilin Jiao: Writing – original draft, Visualization, Methodology, Formal analysis, Data curation. **Xu Zheng:** Writing – review & editing, Validation, Supervision, Methodology, Funding acquisition.

Author statement

We declare that this manuscript is original, has not been published before, and is not currently being considered for publication elsewhere.

We confirm that the manuscript has been read and approved by all named authors, and there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us.

We understand that the Corresponding Author is the primary contacts for the Editorial process. She is responsible for communicating with the other authors about progress, submissions of revisions, and final approval of proofs.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT in order to improve language and readability. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could appear to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.pacfin.2025.102996.

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